

Ch-3 — Heat

Science — Physics · Class 7 · Max Marks: 25 · Time: 45 min

Name: _____ Roll No: _____ Date: _____

Topics covered: Hot & cold and temperature, measuring temperature, clinical and laboratory thermometers, transfer of heat (conduction, convection, radiation), conductors & insulators, sea & land breeze, dark/light surfaces.

Section A — Multiple Choice Questions (1 × 8 = 8)

1. The unit in which temperature is usually measured on a thermometer is the:

- A. metre
- B. kilogram
- C. second
- D. degree Celsius (°C)

2. The normal temperature of a healthy human body is about:

- A. 35°C
- B. 37°C
- C. 100°C
- D. 42°C

3. The temperature range of a clinical thermometer is about:

- A. 35°C to 42°C
- B. -10°C to 110°C
- C. 0°C to 250°C
- D. 0°C to 100°C

4. Heat always flows on its own from a body at ____ temperature to a body at ____ temperature.

- A. zero, high
- B. lower, higher
- C. higher, lower
- D. equal, equal

5. Which of the following is the best conductor of heat?

- A. plastic
- B. wood
- C. rubber
- D. aluminium

6. The other end of a steel spoon kept in a cup of hot tea becomes hot by the process of:

- A. conduction
- B. radiation
- C. evaporation
- D. convection

7. A sea breeze blows from the sea towards the land during the:

- A. midnight
- B. night
- C. day
- D. early hours

8. The heat of the Sun reaches the Earth through empty space mainly by:

- A. conduction
- B. radiation
- C. all three
- D. convection

Section B — Fill in the Blanks (1 × 5 = 5)

1. The device used to measure temperature is a _____.
2. A clinical thermometer has a _____ near the bulb that stops the mercury falling on its own.
3. Materials that do not allow heat to pass through them easily are called _____.
4. In _____, the particles of the medium do not move from their place while passing on heat.
5. Dark-coloured surfaces are _____ absorbers of heat than light-coloured surfaces.

Section C — True or False (1 × 5 = 5)

1. Our sense of touch is a reliable way to measure the exact temperature of an object. (True / False)
2. Water is a poor conductor of heat. (True / False)
3. Convection can take place in solids. (True / False)
4. Woollen clothes keep us warm because the trapped air is a poor conductor of heat. (True / False)
5. A laboratory thermometer is used to measure the temperature of the human body. (True / False)

Section D — Match the Columns (1 × 4 = 4)

Column A	Column B
1. Clinical thermometer	a) a good conductor of heat
2. Land breeze	b) reduces heat transfer by all three modes
3. Copper	c) measures body temperature
4. Thermos flask	d) blows from land to sea at night

Section E — Assertion–Reason (1 × 3 = 3)

Choose: (A) both A and R true and R explains A; (B) both true but R does not explain A; (C) A true, R false; (D) A false, R true.

1. **Assertion (A):** We wear light-coloured clothes in summer.
Reason (R): Light-coloured surfaces reflect most of the heat radiation falling on them.
2. **Assertion (A):** The handles of cooking pans are made of plastic or wood.
Reason (R): Plastic and wood are good conductors of heat.
3. **Assertion (A):** A sea breeze blows during the day.
Reason (R): During the day the land heats up faster than the sea, so the warm air over the land rises.

Answer Key

Multiple Choice Questions

- | | | | | |
|------|------|------|------|------|
| 1. D | 2. B | 3. A | 4. C | 5. D |
| 6. A | 7. C | 8. B | | |

Fill in the Blanks

1. thermometer
2. kink (constriction)
3. insulators (poor conductors)
4. conduction
5. better (good)

True or False

1. False
2. True
3. False
4. True
5. False

Match the Columns

1-c, 2-d, 3-a, 4-b

Assertion–Reason

1. A
2. C

3. A